

SMOKE TEST CASE DOCUMENT

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Version: 1.0

Test Type: Smoke Testing — Critical Path Validation

Smoke Test — Purpose & Execution Guidelines

What is Smoke Testing?

Smoke Testing (also called Build Verification Testing) is a shallow, wide test pass that confirms the most critical end-to-end paths of the EDC Lab Management module are operational before full regression is executed. If any smoke test fails, the build is rejected and returned to development.

Priority Legend

P1 - Critical	System launch blocker — must pass before any testing proceeds
P2 - High	Core functional path — failure severely limits test scope
P3 - Medium	Supporting feature — failure impacts specific workflows only

Smoke Test Suite — Coverage Summary

Module Code	Smoke Test Area	TC Count	Priority
SM-PRM	Lab Permissions — Role Access Verification	4	P1-Critical
SM-LFM	Lab Form Creation & Lab Use Type Assignment	4	P1-Critical
SM-LVM	Lab Variable Settings (Lab Test Result & Determinant)	4	P1-Critical
SM-LAB	Lab Creation, Details & Module Navigation	4	P1-Critical
SM-RRG	Reference Range Setup & Numeric Validation	5	P1-Critical
SM-DET	Range Determinant Configuration	3	P2-High
SM-EFD	Effective Dates — Individual Entry	3	P2-High
SM-LDE	Lab Data Entry — Lab Selection & Save	5	P1-Critical
SM-OOR	Out-of-Range Flag — Trigger & Resolution	5	P1-Critical
SM-FIP	Field Information Popup Accuracy	3	P2-High
SM-UPL	Lab Data Upload — Bulk Upload Mapping	3	P2-High
SM-EXP	Lab Data Export — Export Labs & Include Labs	4	P2-High
SM-AUD	Audit Logs — Download & Column Verification	3	P2-High
SM-DRL	Deactivate & Reactivate Labs	3	P2-High
SM-LCU	LOINC Codes & Units Entry	2	P3-Medium
SM-CPY	Copy Reference Range	2	P3-Medium
SM-NEG	Unsupported Features — DDE/API Block Verification	3	P2-High
SM-MIG	DEV → TEST Deployment Verification	2	P2-High
TOTAL SMOKE TEST CASES		62	

• Validates that Lab permissions section is visible and that the three core permission types enforce correct access boundaries.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-PRM-01	P1-Critical	Labs permission section appears under Configure → Roles & Permissions when Labs is enabled in EDC Admin	1. Ensure Labs checkbox is checked under EDC Admin → Enable. 2. Navigate to Configure → Roles & Permissions. 3. Click on any existing role → PERMISSIONS tab. 4. Scroll to the Labs section.	The 'Labs' section is visible under the PERMISSIONS tab with four sub-permissions: Add/Edit Labs, Deactivate Labs, Read-Only Labs, and Add/Edit Reference Ranges. No error or missing section.	Passed
SM-PRM-02	P1-Critical	User with Add/Edit Labs permission can successfully create a new lab in Lab Management	1. Assign 'Add/Edit Labs' permission to role 'QA_Editor'. 2. Log in as a user with QA_Editor role. 3. Navigate to Manage → Lab Management. 4. Enter lab name 'Smoke Lab A' in the Create Labs field. 5. Click Create.	'Smoke Lab A' appears in the active labs list on the left panel. Labs counter increments to reflect the new lab. No permission-denied error occurs.	Passed
SM-PRM-03	P1-Critical	User with Read-Only Labs permission cannot edit lab details	1. Assign only 'Read-Only Labs' to role 'QA_ReadOnly'. 2. Log in with QA_ReadOnly role. 3. Navigate to Manage → Lab Management → click any lab → DETAILS tab. 4. Attempt to edit the Lab Name field. 5. Attempt to click Save.	All DETAILS fields are read-only — not editable. Save button is either hidden or disabled. No data can be modified. No permission error thrown when viewing, only when attempting to write.	Passed
SM-PRM-04	P2-High	Labs section is hidden when Labs flag is disabled under EDC Admin	1. Navigate to EDC Admin → Enable. 2. Uncheck the 'Labs' checkbox and save. 3. Navigate to Configure → Roles & Permissions → any role → PERMISSIONS tab.	The 'Labs' section does NOT appear in the PERMISSIONS tab. Lab Management is inaccessible from the Manage menu. Re-enabling the checkbox restores both.	Passed

SM-LFM Lab Form Creation & Lab Use Type Assignment

• Verifies Lab form creation via Configure → Forms, correct use-type assignment, and visit selection.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-LFM-01	P1-Critical	Create a new Lab form by selecting 'Lab' as form use type under Configure → Forms	1. Navigate to Configure → Forms. 2. Click 'Add / Import / Copy a Form'. 3. Select form 'Hematology Panel' from the dropdown. 4. In '--Select a form use--', choose 'Lab'. 5. Click Continue. 6. Select visits: Screening and Week 4. 7. Click Add.	'Hematology Panel' is created with form use = Lab. It appears in the forms list with a Lab tag. The selected visits (Screening, Week 4) are associated. No error during creation.	Passed
SM-LFM-02	P1-Critical	Lab form created via Configure → Forms appears under Lab Management LAB TESTS tab	1. Ensure 'Hematology Panel' Lab form exists. 2. Navigate to Manage → Lab Management → click any active lab → LAB TESTS tab. 3. Inspect the form list.	'Hematology Panel' is listed under the LAB TESTS tab with a selectable checkbox. Selecting it and clicking 'Manage Ranges' opens the range editor for that form.	Passed
SM-LFM-03	P1-Critical	Randomization form cannot be assigned 'Lab' form use type	1. Navigate to Configure → Forms → Add / Import / Copy a Form. 2. Select the Randomization form. 3. In '--Select a form use--', check whether 'Lab' is available.	The 'Lab' option is either absent or disabled for the Randomization form type. System prevents saving a Randomization form as a Lab form.	Passed
SM-LFM-04	P2-High	Out-of-Range flag visible in EDC view but absent in DDC view for the same Lab form data	1. Open a Lab form for a subject in DDC mode. 2. Enter a lab test result value that exceeds the configured upper range limit. 3. Save in DDC. 4. Open the same form for the same subject in EDC view.	DDC view: No out-of-range flag icon appears next to the variable. EDC view: The higher out-of-range flag icon appears next to the variable, confirming flags are EDC-only.	Not Executed

SM-LVM Lab Variable Settings — Lab Test Result & Range Determinant

i Confirms correct variable type settings, PHI restriction, and single-setting constraint.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-LVM-01	P1-Critical	Number variable set as Lab Test Result appears in Lab Management Manage Ranges	1. Navigate to Configure → Forms → 'Hematology Panel'. 2. Add a Number variable 'WBC Count'. 3. In Variable Editor, under Lab Data, select 'Lab Test Result'. 4. Save and Close. 5. Navigate to Manage → Lab Management → select a lab → LAB TESTS → check Hematology Panel → Manage Ranges.	'WBC Count' appears in the Manage Ranges view with an edit icon. Lower Limit and Upper Limit fields are available for configuration. No error on save.	Passed
SM-LVM-02	P1-Critical	Categorical Select variable set as Range Determinant is available in reference range editor	1. On any form, add a Categorical Select variable 'Gender' (options: Male, Female). 2. In Variable Editor, under Lab Range, select 'Range Determinant'. 3. Save and Close. 4. Open reference range editor in Lab Management → click Range Determinant field.	'Gender' variable is listed in the Range Determinant dropdown with its values (Male, Female) selectable. It can be used to stratify reference ranges.	Passed
SM-LVM-03	P1-Critical	Lab Test Result variable cannot be marked as PHI	1. Open 'WBC Count' Variable Editor. 2. Ensure 'Lab Test Result' is set. 3. Attempt to enable the PHI checkbox. 4. Click Save.	System blocks save with validation message: 'Lab Test Result variables cannot be marked as PHI'. PHI flag is either disabled or triggers an error when combined with Lab Test Result setting.	Passed
SM-LVM-04	P1-Critical	Same variable cannot have both Lab Test Result and Range Determinant settings simultaneously	1. Open a Number variable in Variable Editor. 2. Set Lab Data → Lab Test Result. 3. Save. 4. Reopen Variable Editor. 5. Attempt to also set Lab Range → Range Determinant on the same variable.	System prevents setting both options on one variable. When Lab Test Result is active, the Range Determinant option is grayed out or blocked. Validation error confirms only one Lab setting is permitted per variable.	Passed

SM-LAB Lab Creation, Details & Module Navigation

i Verifies lab creation, detail entry, Labs counter accuracy, and module UI elements.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-LAB-01	P1-Critical	Create a lab via Manage → Lab Management and verify it appears in the active labs list	1. Navigate to Manage → Lab Management. 2. In the Create Labs form, type 'BioTest Laboratory'. 3. Click Create.	'BioTest Laboratory' appears in the left panel under Labs. The Labs (#) counter reflects the correct active lab count. DETAILS and LAB TESTS tabs appear. Three-dot icon is visible next to the lab name.	N/E
SM-LAB-02	P1-Critical	Lab details (name, lab code, address) can be entered and saved	1. Click 'BioTest Laboratory' in the labs list → DETAILS tab. 2. Enter Lab Code: 'BTL-001'. 3. Enter Address: '500 Science Drive, Boston, MA 02115'. 4. Click Save.	Lab code 'BTL-001' and address '500 Science Drive, Boston, MA 02115' are saved and visible when the DETAILS tab is revisited. Labs list reflects the saved lab name.	N/E
SM-LAB-03	P1-Critical	Labs (#) counter shows only active labs count	1. Note current Labs (#) value with 3 active and 1 deactivated lab. 2. Verify the number displayed in the label.	Labs (#) displays '3' — counting only active labs. Deactivated lab is excluded from the counter. Counter is accurate.	N/E
SM-LAB-04	P2-High	Export Labs generates a downloadable Excel file with lab details	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to Save button. 3. Click 'Export Labs'. 4. Open the downloaded file.	An Excel (.xlsx) file downloads successfully. File contains a tab for lab details and individual tabs for each lab's reference ranges. File opens without corruption.	N/E

SM-RRG Reference Range Setup & Numeric Validation

Validates numeric range creation, default values, blank-field validation, and range name uniqueness.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-RRG-01	P1-Critical	Reference range with valid lower and upper numeric limits can be created and saved	1. Navigate to Manage → Lab Management → BioTest Laboratory → LAB TESTS. 2. Check 'Hematology Panel' → click Manage Ranges. 3. Click edit icon next to 'WBC Count'. 4. Enter Name: 'WBC Adult Normal'. 5. Enter Lower Limit: 4.5, Upper Limit: 11.0. 6. Click Save icon.	Reference range 'WBC Adult Normal' is saved with Lower = 4.5 and Upper = 11.0. It appears under WBC Count in the Manage Ranges view. No error on save.	N/E
SM-RRG-02	P1-Critical	Blank numeric range limits trigger 'Invalid Entry' validation error	1. Open reference range editor for 'WBC Count'. 2. Enter Name: 'Blank Test Range'. 3. Clear both Lower Limit and Upper Limit fields. 4. Click Save icon.	System displays 'Invalid Entry. Hover for Details' on both empty fields. Save is blocked. Fields are highlighted. Default value of 0 reappears when editor is re-opened.	N/E
SM-RRG-03	P1-Critical	Default lower and upper limit values are pre-set to 0 on a new range	1. Open reference range editor for a numeric Lab Test Result variable. 2. Click the + icon to create a new reference range. 3. Observe the Lower Limit and Upper Limit fields before entering any data.	Both Lower Limit and Upper Limit fields show '0' as the default value — not blank. System pre-populates 0 to prevent accidental blank saves.	N/E
SM-RRG-04	P1-Critical	Reference range name must be unique — duplicate name is rejected	1. Ensure reference range 'WBC Adult Normal' already exists for WBC Count. 2. Click + icon to add another range. 3. Enter Name: 'WBC Adult Normal' (duplicate). 4. Click Save icon.	System blocks save with validation: 'Range name must be unique for this variable'. Save is prevented until a different name is entered.	N/E
SM-RRG-05	P1-Critical	Categorical Yes/No Lab Test Result variable allows Yes or No as range value	1. Add a Categorical Yes/No variable 'X-Ray Taken' as Lab Test Result to a Lab form. 2. In reference range editor, create range 'X-Ray Expected'. 3. In range values section, select 'Yes'. 4. Click Save.	Range 'X-Ray Expected' is saved with expected answer = Yes. It appears in the Manage Ranges view. During data entry, entering 'No' will trigger an out-of-range flag.	N/E

SM-DET Range Determinant Configuration

Verifies determinant variables are selectable and correctly stratify reference range flag behavior.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-DET-01	P2-High	Gender Range Determinant variable appears in reference range editor and is selectable	1. Ensure 'Gender' (Categorical Select) is set as Range Determinant. 2. Open reference range editor for 'WBC Count'. 3. Click the Range Determinant field.	'Gender' with its values (Male, Female) is listed in the Range Determinant dropdown. It can be selected and a specific value (e.g., Female) can be set for the range stratification.	N/E
SM-DET-02	P2-High	Out-of-range flag triggers only when the Range Determinant condition is met	1. Create 'WBC Female Range' with Gender=Female, Lower=4.0, Upper=10.5. 2. Subject S-001 has Gender=Female recorded. 3. Open CBC Panel for S-001, select lab, enter WBC=12.0. 4. Save.	Higher out-of-range flag appears next to WBC Count (12.0 exceeds female upper limit 10.5). The range fires because the Gender=Female determinant condition is satisfied.	N/E
SM-DET-03	P2-High	No out-of-range flag when Range Determinant data is missing for subject	1. Subject S-002 has NO Gender value recorded. 2. 'WBC Female Range' exists with Gender=Female determinant. 3. Enter WBC=15.0 for S-002 and save.	No out-of-range flag appears. Without the Gender determinant value filled in, the range condition is unmet and flag evaluation is skipped regardless of the WBC value.	N/E

SM-EFD Effective Dates — Individual Entry

Confirms effective date entry, date format enforcement, and start-before-end validation.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-EFD-01	P2-High	Effective dates (Start and End) can be entered and saved for a reference range	1. Open reference range 'WBC Adult Normal'. 2. In Effective Dates section, enter Start Date: 01/01/2025. 3. Enter End Date: 12/31/2025. 4. Click Save icon.	Effective dates 01/01/2025 – 12/31/2025 are saved. The dates appear below the range name. A Calendar icon appears next to the WBC Count lab test name in the LAB TESTS tab.	N/E
SM-EFD-02	P2-High	System rejects Start Date that occurs after End Date	1. Open effective dates for any reference range. 2. Enter Start Date: 12/31/2025. 3. Enter End Date: 01/01/2025. 4. Click Save.	System displays validation error: 'Start date must occur before the end date'. Save is blocked. Fields are highlighted to indicate invalid date range.	N/E
SM-EFD-03	P2-High	Effective dates are optional — reference range saves successfully with blank date fields	1. Create a new reference range 'RBC Normal Adult' with valid lower/upper limits. 2. Leave Start Date and End Date fields empty. 3. Click Save.	Reference range saves successfully without effective dates. No validation error. Calendar icon does NOT appear next to the lab test name.	N/E

SM-LDE Lab Data Entry — Lab Selection & Save

Validates the core data entry flow: lab field population, mandatory lab selection, and reference range display after save.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-LDE-01	P1-Critical	Lab dropdown in a Lab form is populated with active labs only	1. Navigate to a subject's Hematology Panel Lab form. 2. Click the Lab field at the top of the form.	The Lab dropdown lists all active labs (e.g., BioTest Laboratory, Quest Diagnostics). Deactivated labs do NOT appear in the dropdown. Search within the field narrows results as the user types.	N/E
SM-LDE-02	P1-Critical	Lab form cannot be saved without selecting a lab	1. Open Hematology Panel for a new subject. 2. Enter WBC Count = 6.5. 3. Leave the Lab field blank. 4. Click Save.	System blocks save with validation message: 'A lab must be selected to save this lab form'. All entered data is retained in the form. Lab field is highlighted as required.	N/E
SM-LDE-03	P1-Critical	Reference range displays as read-only below the variable after lab selection and save	1. Open Hematology Panel for subject S-003. 2. Select 'BioTest Laboratory' in the Lab field. 3. Enter WBC Count = 7.0. 4. Click Save.	After save, reference range '4.5 – 11.0' appears below the WBC Count field as read-only text. The range is derived from BioTest Laboratory's configuration and cannot be edited on the form.	N/E
SM-LDE-04	P1-Critical	Data entry user can view out-of-range flags without special Lab permissions	1. Log in as a standard data entry user (no specific Lab permission). 2. Open a Lab form where a variable has an active out-of-range flag. 3. View the flag.	Out-of-range flag icon is visible to the data entry user without any additional permissions. By default, all users with Lab form access can view flags.	N/E
SM-LDE-05	P1-Critical	Correct wrong lab using the Change Lab button	1. Subject S-004 has 'BioTest Laboratory' saved on Hematology Panel. 2. Open the form. 3. Change Lab field to 'Quest Diagnostics'. 4. Click the 'Change Lab' button next to Save & Next.	'Change Lab' button is visible after initial save. After clicking it, lab updates to Quest Diagnostics. Reference ranges refresh to Quest Diagnostics ranges. The form reflects the new lab immediately.	N/E

SM-OOR Out-of-Range Flag — Trigger & Resolution

i Core smoke path: flag appearance for above/below range, boundary value handling, and flag resolution on correction.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-OOR-01	P1-Critical	Higher out-of-range flag appears when value exceeds the upper limit	- For subject S-005, open Hematology Panel. - Select BioTest Laboratory (WBC range: 4.5–11.0). - Enter WBC Count = 14.0. - Click Save.	Higher out-of-range flag icon appears immediately next to WBC Count after save. Reference range '4.5 – 11.0' is displayed below the field. Flag icon is visually distinct (upward indicator).	N/E
SM-OOR-02	P1-Critical	Lower out-of-range flag appears when value is below the lower limit	- For subject S-006, enter WBC Count = 2.0 (below lower limit 4.5). - Save.	Lower out-of-range flag icon appears next to WBC Count. Icon is visually distinct from the higher flag (downward indicator). Reference range is displayed.	N/E
SM-OOR-03	P1-Critical	No out-of-range flag for value exactly at the lower boundary	- Enter WBC Count = 4.5 (equal to lower limit). - Save.	No out-of-range flag appears. Value 4.5 is treated as within range (inclusive boundary). Reference range is shown below the field without a flag icon.	N/E
SM-OOR-04	P1-Critical	Out-of-range flag resolves to Inactive when corrected value is within range	- Subject S-005 has WBC = 14.0 with an active higher flag. - Update WBC Count to 7.0. - Save. - Click the former flag area to open Field Information popup.	Higher flag icon disappears from the form. Field Information popup shows Status = Inactive. The correction of value within range deactivates the flag without deleting the audit record.	N/E
SM-OOR-05	P1-Critical	Out-of-range flags remain on historical data after lab deactivation	- Subject S-007 has WBC = 15.0 with an active flag under BioTest Laboratory. - Deactivate BioTest Laboratory. - Reopen Hematology Panel for S-007.	Out-of-range flag REMAINS on S-007's form after lab deactivation. Historical flags are preserved. Lab deactivation does not retroactively remove flags from existing subject data.	N/E

SM-FIP Field Information Popup Accuracy

i Validates the five required popup fields and status transition from Active to Inactive.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-FIP-01	P2-High	Field Information popup displays all five required data points for an active out-of-range flag	- Subject S-008 has WBC = 14.0 flagged under BioTest Laboratory, range 'WBC Adult Normal' (4.5–11.0). - Click the out-of-range flag icon next to WBC Count.	Popup displays all five data points: 1. Range Name: WBC Adult Normal 2. Expected Range: 4.5 – 11.0 3. Lab Name: BioTest Laboratory 4. Flag Status: Active 5. Timestamp: (date and time flag was triggered)	N/E
SM-FIP-02	P2-High	Popup status updates to Inactive when variable value is corrected within range	- Update WBC Count for S-008 from 14.0 to 8.0. - Save. - Click the flag area to open Field Information popup.	Popup shows Flag Status = Inactive. Flag icon is no longer visible on the form. Popup retains the original activation timestamp and the range details for audit reference.	N/E
SM-FIP-03	P2-High	Field Information popup is accessible to standard data entry users	- Log in as standard data entry user. - Open a Lab form with an active out-of-range flag. - Click the flag icon.	Field Information popup opens successfully without needing Lab-specific permissions. The popup content (range name, expected range, lab name, status, timestamp) is fully visible.	N/E

SM-UPL Lab Data Upload — Bulk Upload Mapping

✓ Confirms Lab field is present in bulk upload mapping and data uploads trigger correct range evaluation.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-UPL-01	P2-High	Lab field appears as a required mapping field in Bulk Upload for a Lab form	- Navigate to Configure → Bulk Upload → Add new field mapping. - Select 'Hematology Panel' as the form. - Select a valid upload file. - Inspect the Form (Field) Name column in the mapping table.	'Lab' appears as a field in the Form (Field) Name column and must be mapped before the upload can be saved. It is treated as a required field for Lab forms.	N/E
SM-UPL-02	P2-High	Bulk upload with correct lab mapping populates subjects' forms with the correct lab	- Map the 'LabName' column from the upload file to the 'Lab' field. - Complete all other required field mappings. - Upload a file with BioTest Laboratory as the lab for 5 subjects. - Open CBC Panel for any of the 5 uploaded subjects.	The Lab field on each subject's form shows 'BioTest Laboratory'. Data is populated correctly. No upload failure or missing lab assignment.	N/E
SM-UPL-03	P2-High	Out-of-range flags trigger for values uploaded via Bulk Upload	- Upload file contains WBC Count = 16.0 for subject S-010 in BioTest Laboratory (range 4.5–11.0). - After upload, open Hematology Panel for S-010.	Higher out-of-range flag is visible next to WBC Count (16.0 exceeds upper limit 11.0). Bulk uploaded data is subject to the same range evaluation as manually entered data.	N/E

SM-EXP Lab Data Export — Export Labs & Data Export with Include Labs

✓ Validates Export Labs and Data Export (Include Labs) outputs.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-EXP-01	P2-High	Export Labs generates Excel with lab details and reference range data	- Navigate to Manage → Lab Management. - Click three-dot icon next to Save → 'Export Labs'. - Open the downloaded Excel file.	Excel file downloads without error. Contains lab details (name, code, address) in one tab and individual reference range tabs per lab. LOINC codes and units are included where configured.	N/E
SM-EXP-02	P2-High	Data Export with Include Labs includes all required lab-specific columns	- Navigate to Analysis → Data Export. - Select Hematology Panel form. - Check 'Include Labs'. - Click Download. - Open file and inspect column headers.	Export file includes: Lab Name, Flag State, Reference Range values, Range Determinant values, Out-of-Range (Within/Above/Below/Equal/Not Equal), Lower Limit, Upper Limit, Expected Value. All required columns present.	N/E
SM-EXP-03	P2-High	Data Export without Include Labs does not contain lab-specific fields	- Navigate to Analysis → Data Export. - Select Hematology Panel. - Do NOT check 'Include Labs'. - Download and inspect columns.	Export does not contain Lab Name, Flag State, Out-of-Range, Reference Range, or any Lab Management-specific columns. Standard clinical data columns only.	N/E
SM-EXP-04	P2-High	Flag State values in Data Export match correct status codes	- Export with Include Labs for subjects with WBC values: within range, above range, below range. - Inspect the 'Out-of-Range' column values.	Out-of-Range column contains exactly one of: Within, Above, Below, Equal, Not Equal — with no other values present. Each subject row reflects the correct flag state.	N/E

SM-AUD Audit Logs — Download & Column Verification

i Verifies both audit logs download correctly and contain all required tracking columns.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-AUD-01	P2-High	Lab Details Audit Log downloads as Excel and contains all required columns	1. Make a lab detail change (e.g., update lab address). 2. Navigate to three-dot icon next to Save → 'Lab Details Audit Log'. 3. Download and open the file. 4. Verify column headers.	Excel file downloads successfully. Contains all required columns: Lab Name, Object, Action, Old Value, New Value, Date, Time, Changed By. The address change is reflected with correct Old and New values.	N/E
SM-AUD-02	P2-High	Reference Range Audit Log captures range value changes with full context	1. Change WBC Adult Normal upper limit from 11.0 to 12.0. 2. Download Reference Range Audit Log. 3. Locate the change entry.	Audit log entry shows: Lab Form = Hematology Panel, Lab Test Name = WBC Count, Range Name = WBC Adult Normal, Object = Upper Limit, Action = Updated, Old Value = 11.0, New Value = 12.0, Changed By = (username), with correct timestamp.	N/E
SM-AUD-03	P2-High	Reference Range Audit Log captures effective date additions	1. Add effective dates 01/01/2025 – 12/31/2025 to 'WBC Adult Normal'. 2. Download Reference Range Audit Log. 3. Find the effective date entries.	Two audit rows: one for Start Date Added (New Value = 01/01/2025) and one for End Date Added (New Value = 12/31/2025). Both entries reference WBC Adult Normal range and have correct timestamps.	N/E

SM-DRL Deactivate & Reactivate Labs

i Validates deactivation removes labs from data entry and reactivation fully restores them.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-DRL-01	P2-High	Deactivate a lab and verify it is removed from data entry Lab dropdown	1. Navigate to Manage → Lab Management. 2. Click three-dot icon next to 'BioTest Laboratory'. 3. Click 'Deactivate Lab'. 4. Open any Lab form for a subject → click Lab field.	'BioTest Laboratory' no longer appears in the Lab field dropdown. Labs counter decrements. Lab still visible under 'Deactivated Labs' from the three-dot menu.	N/E
SM-DRL-02	P2-High	Reactivate a deactivated lab and verify it reappears in data entry	1. Navigate to Manage → Lab Management → three-dot icon → Deactivated Labs. 2. Click three-dot icon next to 'BioTest Laboratory' → Reactivate Lab. 3. Open a Lab form for a subject → click Lab field.	'BioTest Laboratory' reappears in the active Labs list. Labs counter increments. Lab field dropdown in data entry includes 'BioTest Laboratory' again. All previously configured details and ranges are intact.	N/E
SM-DRL-03	P2-High	Out-of-range flags remain on subject data after lab deactivation	1. Subject S-011 has an active out-of-range flag from BioTest Laboratory. 2. Deactivate BioTest Laboratory. 3. Open Hematology Panel for S-011.	Historical out-of-range flag is still visible on S-011's form. Deactivating a lab does not retroactively clear flags from subject records. The Field Information popup still shows the lab name.	N/E

SM-LCU LOINC Codes & Units Entry

i Confirms LOINC codes and units can be saved and appear in exports.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-LCU-01	P3-Medium	LOINC code and unit can be saved for a reference range	1. Navigate to Manage → Lab Management → BioTest Laboratory. 2. Click reference range 'WBC Adult Normal'. 3. Enter Lab Test Unit: 'K/uL'. 4. Enter LONIC: '26464-8'. 5. Click Save icon.	LOINC code '26464-8' and unit 'K/uL' are saved without error. They are visible in the reference range view. Values appear in the Data Export and Export Labs Excel files.	N/E

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-LCU-02	P3-Medium	LOINC code and unit appear in Export Labs Excel output	1. Ensure LOINC '26464-8' and unit 'K/uL' are saved for WBC Count. 2. Export Labs. 3. Open Excel → locate WBC Count row.	Export file includes 'K/uL' in the Unit column and '26464-8' in the LOINC column for the WBC Count reference range row.	N/E

SM-CPY Copy Reference Range

i Confirms copy preserves all attributes and stays within the same lab.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-CPY-01	P3-Medium	Copy a reference range and verify all attributes are preserved in the copy	1. Navigate to Manage → Lab Management → BioTest Laboratory → WBC Adult Normal. 2. Click three-dot icon next to range name → 'Copy Reference Range'. 3. Open the copied range.	A new range 'WBC Adult Normal (Copy)' or similar is created. All attributes match the source: lower = 4.5, upper = 11.0, effective dates, LOINC, and units. Copied range is listed under the same lab only.	N/E
SM-CPY-02	P3-Medium	Copied range stays within BioTest Laboratory and does not appear in other labs	1. Copy 'WBC Adult Normal' in BioTest Laboratory. 2. Navigate to Quest Diagnostics → LAB TESTS → Manage Ranges for Hematology Panel.	The copied range does NOT appear in Quest Diagnostics. Copy is isolated to the source lab (BioTest Laboratory) only.	N/E

SM-NEG Unsupported Features — DDE / API Block Verification

i Confirms that known unsupported features are correctly blocked for Lab forms.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-NEG-01	P2-High	Double Data Entry (DDE) is not supported for Lab forms	1. Attempt to enable or initiate Double Data Entry workflow for a Lab form (e.g., Hematology Panel). 2. Observe system behavior.	DDE is not available for Lab forms. The option is either absent, grayed out, or a restriction message is shown: 'Lab Management does not support Double Data Entry'.	N/E
SM-NEG-02	P2-High	Labs section is inaccessible when EDC Admin Labs flag is disabled	1. Disable Labs under EDC Admin → Enable. 2. Navigate to Manage menu. 3. Navigate to Configure → Roles & Permissions → any role → PERMISSIONS tab.	Manage menu does not show 'Lab Management'. PERMISSIONS tab does not show 'Labs' section. Both are fully hidden when the EDC Admin Labs flag is off.	N/E
SM-NEG-03	P2-High	Import API does not process Lab field from Lab form submissions	1. Attempt to submit lab form data including a Lab field value via the Import API. 2. Inspect the API response or resulting data in the study.	Import API does not process the Lab field. Either the call fails with an appropriate error or the Lab field is ignored. Lab Management data must be entered via UI-based workflows.	N/E

SM-MIG DEV → TEST Deployment Verification

i Verifies that Lab configuration (forms, variables, permissions) deploys correctly from DEV to TEST.

TC-ID	Priority	Smoke Test Title	Execution Steps	Expected Result	Status
SM-MIG-01	P2-High	Lab form and Lab variable settings are preserved after DEV → TEST deployment	1. In DEV: confirm Hematology Panel is a Lab form and WBC Count is set as Lab Test Result. 2. Deploy DEV configuration to TEST. 3. In TEST: open Configure → Forms → Hematology Panel → Variable Editor for WBC Count.	In TEST: Hematology Panel retains Lab form use type. WBC Count retains Lab Test Result setting. Gender retains Range Determinant setting. All Lab variable configurations are preserved post-deployment.	N/E
SM-MIG-02	P2-High	Lab permissions for roles are preserved after DEV → TEST deployment	1. In DEV: role 'QA_Editor' has Add/Edit Labs permission. 2. Deploy to TEST. 3. In TEST: Configure → Roles & Permissions → QA_Editor → PERMISSIONS tab.	QA_Editor role in TEST has Add/Edit Labs permission intact. Permissions are not lost during environment deployment.	N/E